Computers & Graphics 132 (2025) 104366

Contents lists available at ScienceDirect

Computers & Graphics

journal homepage: www.elsevier.com/locate/cag

Special Section on CAD/Graphics 2025

Hyperspectral image denoising via total generalized variation regularized low-rank tensor decomposition

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ARTICLE INFO

Keywords:
Hyperspectral image
Total generalized variation
Low-rank tensor decomposition
Detail-preserving

ABSTRACT

Recent studies have demonstrated that total generalized variation (TGV) effectively preserves image features while avoiding staircase artifacts common in total variation (TV) regularization. This paper proposes a novel hyperspectral image (HSI) denoising method that extends conventional 2D TGV to 3D TGV in the spatial–spectral domain through a tensor-formulated framework, where first-order term characterizes spatial–spectral gradients and second-order term encodes both spatial/spectral curvature and cross-band curvature. Specifically, the proposed approach synergistically combines this 3D TGV regularization, maintaining an adaptive balance between detail preservation and smooth transition restoration, with Tucker decomposition that captures global spectral low-rankness. To solve the resulting optimization problem, we develop an efficient iterative algorithm based on the alternating direction method of multipliers (ADMM). Extensive results and comparisons on simulated and real HSIs with mixed noise demonstrate that our method achieves superior denoising performance, yielding significant improvements in both visual quality and quantitative metrics (e.g., MPSNR, MSSIM, ERGAS) compared to existing approaches.

1. Introduction

HSIs provide rich spatial and spectral information and are widely used in environmental monitoring [1], military surveillance [2], mineral exploration [3], and more. However, due to the limitations of imaging equipment and environmental factors, HSIs are inevitably affected by various types of noise during acquisition and transmission, including Gaussian noise, impulse noise, stripes, and deadlines. These noises severely degrade the visual quality and data accuracy of HSIs, thereby impacting the performance of subsequent applications such as target detection [4], segmentation [5], classification [6], and unmixing [7]. Thus, developing noise-robust HSI restoration methods that maintain spatial–spectral fidelity remains a critical challenge in HSI processing.

Existing methods can be broadly categorized into model-driven and data-driven methods (e.g., deep learning). While deep learning methods [8–10] attempt to directly learn a denoiser from paired noisy-clean HSIs in an end-to-end way. Although they can achieve promising denoising results, the performance of these methods heavily depends on the quality of the training data and lacks generalization ability. In contrast, model-driven methods typically leverage the following prior information of HSIs: nonlocal self-similarity [11–14], low- rankness [15–18], and piecewise smoothness [19–22].

Among them, the low-rankness is a classic global correlation property that reflects the inherent redundancy within tensors. Common low-rank tensor decompositions include PARAFAC (CP) decomposition [23], Tucker decomposition [24], tensor network decomposition [25], and tubal rank based on tensor singular value decomposition (t-SVD) [26]. Additionally, piecewise smoothness is another important prior, reflecting the local continuity of HSIs in the spatial–spectral dimension. The current mainstream model-driven HSI denoising methods combine the two priors of low-rankness and piecewise smoothness. TV regularization has emerged as a pivotal mathematical framework for characterizing piecewise smoothness. Early methods typically flatten HSIs along the spectral dimension into 2D matrices for processing [19, 20], which disrupts the intrinsic structure of the original tensor and neglects spectral correlation. To preserve multidimensional tensor characteristics, tensor-based TV methods have been proposed, including ASSTV [21], SSATV [22], LRTDTV [15], and others. Although these methods preserve local smoothness by jointly constraining spatial and spectral gradients, their reliance solely on first-order gradient information tends to convert smooth transition regions into piecewise constant variations, often introducing staircase artifacts and resulting in the loss of fine details.

To alleviate the undesirable artifacts caused by the TV related regularization, researchers have proposed novel variational methods

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https://doi.org/10.1016/j.cag.2025.104366
Received 27 June 2025; Received in revised form 25 July 2025; Accepted 28 July 2025
Available online 8 August 2025

that combine first-order and higher-order terms, such as H^2DTV-NLRTR [27] and GHSSTV [28]. Although these methods perform better than TV-related methods, they still exhibit artifacts near sharp features, flatten fine details, and fail to adaptively balance detail preservation with smooth transition restoration. Therefore, a key issue that current HSI denoising methods need to address is achieving a good balance between noise removal and detail preservation.

In recent years, the TGV, proposed by Bredies et al. [29], has emerged as an effective solution to overcome the limitations of TV. TGV constructs polynomial combinations of arbitrary orders, automatically balancing first- and higher-order variations to reconstruct piecewise polynomial functions. It can be interpreted as using first-order variations to preserve important image features and details while approximating smooth transition regions effectively through higher-order variations, thereby avoiding staircase effects. TGV has been successfully applied to 2D image processing [30–32]. Recently, Xu et al. [33] introduced tensor robust principal component analysis with TGV for high-dimensional data recovery. However, this approach still relies on traditional 2D TGV and only considers the local spatial smoothness.

HSIs are 3D data cubes containing both spatial and spectral information. Traditional 2D TGV methods process each spectral band independently, ignoring spectral continuity and inducing distortion. In addition, noise in HSIs is typically unevenly distributed across spatial and spectral dimensions, making spatial-only regularization insufficient. Addressing the challenge of designing a computational framework for derivatives spanning from the first to the kth order, we extend TGV to 3D for joint modeling of spatial and spectral features, thereby improving restoration quality. The contributions of our work are as follows:

(1) We extend the 2D TGV to 3D TGV by incorporating gradient operators along all three dimensions. This enables the 3D TGV to model variations in spatial, spectral, and cross-band curvature, thereby preserving higher-order structural details.

(2) By incorporating 3D TGV regularization into the Tucker decomposition framework, we simultaneously exploit the global low-rank property and local spatial–spectral variations, achieving artifact-free reconstruction with preserved fine details and global structures.

(3) The proposed 3D TGV regularized Tucker decomposition model is solved via an ADMM algorithm. Experimental results on both simulated and real data show that the method exhibits a significant performance advantage over several competitive methods, both numerically and visually, especially in the reduction of staircase artifacts and the preservation of details.

The rest of this paper is organized as follows. Section 2 reviews related works. Section 3 introduces the background of Tucker decomposition and TGV, along with some tensor notations. Section 4 proposes the 3D TGV regularized low-rank tensor decomposition model and describes its optimization process. Section 5 conducts experiments and analysis on both simulated and real HSIs data, validating the effectiveness of the proposed method. Finally, Section 6 concludes this work.

2. Related work

In this section, we briefly review related studies on the HSI denoising task, which can be roughly categorized into model-driven methods and data-driven approaches.

2.1. Data-driven methods

Data-driven methods aim to learn denoising mappings directly from data rather than relying on handcrafted priors. In particular, deep neural networks have shown strong performance due to their powerful nonlinear representation capabilities. Various architectures have been explored for HSI denoising, such as 2D/3D convolutional neural network-based [34–37], U-Net-based [38], attention mechanism-based [39], transformer-based [40,41], deep prior-based approaches [42,43], and diffusion model-based methods [44,45]. By learning from large-scale training datasets, these methods can capture complex noise characteristics and spatial–spectral correlations, often achieving promising denoising results. However, their performance may be limited by the restricted training data, especially when generalizing to real noisy HSIs contaminated by unknown or mixed types of noise.

2.2. Model-driven methods

Model-driven methods aim to design effective regularization terms that embed prior information of HSI data into the denoising process. Commonly exploited priors include nonlocal self-similarity, spectral low-rankness and spatial local smoothness. Classic nonlocal self-similarity-based denoising methods include the nonlocal means filter [11], BM3D [12], BM4D [13], nonlocal tensor dictionary learning [46], and intrinsic tensor sparsity regularization [14]. Early methods based on low-rankness primarily utilized low rank matrix models, such as [47–49]. To better preserve the intrinsic multi-dimensional structure of HSIs, recent approaches model HSIs as tensors, effectively leveraging their spatial–spectral redundancy [50–54]. Piecewise smoothness is typically modeled using TV regularization and is often incorporated into low-rank decomposition frameworks to achieve improved restoration performance. Early studies employed band-by-band 2D TV methods, include LRTV [19] and SSTV [20], which disrupted the original tensor structure. To better preserve the intrinsic multi-dimensional structure of HSIs, tensor-based TV methods have been subsequently developed, such as ASSTV [21], SSATV [22], LRTDTV [17], TV-NLRTD [17], E-3DTV [55], 3DCrTV [56] and 3DCTV-RPCA [57]. A commonly known drawback of the TV regularization is that it tends to produce staircase artifacts in smoothly varying regions. To address this issue, higher-order methods [27,28] have been introduced, which can reduce staircase artifacts. However, they still exhibit artifacts near sharp features and flatten fine details. Thus, it is still challenging to preserve details while simultaneously recover smooth transition variations. In this work, we develop the 3D TGV regularization, which not only effectively avoids staircase effects but also adaptively balances the preservation of structural details and the recovery of smooth transitions.

3. Background

3.1. Notations

A tensor, denoted as $\mathcal{X}$, is a multi-dimensional array generalizing scalars (x), vectors ($\boldsymbol{x}$), and matrices ($\boldsymbol{X}$). An Nth order tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$, its mode-$n$ matricization, denoted by $\mathbf{X}_{(n)} \in \mathbb{R}^{I_n\times(\prod_{k\neq n} I_k)}$, is obtained by arranging the mode-n fibers of $\mathcal{X}$ as the columns of a matrix. The multi-linear rank is represented as an array $(r_1, r_2, \dots, r_N)$, where r_n is the rank of $\boldsymbol{X}_{(n)}$, for $n = 1, 2, \dots, N$. The mode-n multiplication of a tensor $\mathcal{X}$ with a matrix $\boldsymbol{U}$ is denoted by $\mathcal{X}\times_n \boldsymbol{U}$, with elements defined as $(\mathcal{X}\times_n \boldsymbol{U})_{i_1,\dots,i_{n-1},j,i_{n+1},\dots,i_N} = \sum_{i_n} x_{i_1,i_2,\dots,i_N} \cdot u_{j,i_n}$. The inner product of two same-sized tensors $\mathcal{X}$ and $\mathcal{Y}$ is the sum of their element-wise products: $\langle\mathcal{X},\mathcal{Y}\rangle = \sum_{i_1,i_2,\dots,i_N} x_{i_1,i_2,\dots,i_N} \cdot y_{i_1,i_2,\dots,i_N}$. The Frobenius norm is defined as: $\|\mathcal{X}\|_F = \sqrt{\langle\mathcal{X},\mathcal{X}\rangle}$, which is the square root of the sum of the squares of all elements in the tensor. The ℓ_1- norm of tensor $\mathcal{X}$ is defined as: $\|\mathcal{X}\|_1 = \sum_{i_1,i_2,\dots,i_N} |x_{i_1,i_2,\dots,i_N}|$, representing the sum of absolute values of all tensor elements.

3.2. Tucker decomposition

Tucker decomposition is a tensor decomposition technique that effectively extracts and utilizes redundant information. HSIs as a form of three-dimensional data cubes, contain rich spatial and spectral information. Tensor decomposition methods can naturally handle this multidimensional data, revealing underlying structures and relationships. The Tucker decomposition of a third-order tensor $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ is expressed as follows:

$$\mathcal{X} \approx \mathcal{G}\times_1 A\times_2 B\times_3 C = \sum_{p=1}^{P}\sum_{q=1}^{Q}\sum_{r=1}^{R} g_{pqr}\mathbf{a}_p\circ\mathbf{b}_q\circ\mathbf{c}_r = [\mathcal{G}; A, B, C],$$

Here, $A \in \mathbb{R}^{I\times P}$, $B \in \mathbb{R}^{J\times Q}$, and $C \in \mathbb{R}^{K\times R}$ are the factor matrices and can be thought of as the principal components in each mode. The tensor $\mathcal{G} \in \mathbb{R}^{P\times Q\times R}$ is called the core tensor, and its entries indicate the level of interaction between the different components.

3.3. Background of TGV

TGV is an advanced regularization technique that extends the classical TV. It is particularly effective in addressing challenges such as noise reduction and edge preservation in image processing. Given a 2D image $\boldsymbol{U}$, the traditional TGV of $\boldsymbol{U}$ is defined as follows:

$$\mathrm{TGV}_\alpha^k(\boldsymbol{U}) = \sup_{v}\bigg\{\int_\Omega \boldsymbol{U}\,\mathrm{div}^k(v)\,dx \mid v\in C_c^k(\Omega,\mathrm{Sym}^k(\mathbb{R}^d)), \|\mathrm{div}^j(v)\|_\infty \le \alpha_j,\ j=0,\dots,k-1\bigg\},$$

where $\alpha = (\alpha_0,\dots,\alpha_{n-1})$ is a fixed positive vector, and $\mathrm{Sym}^k(\mathbb{R}^d)$ denotes the space of symmetric k-tensors on $\mathbb{R}^d$ as

$$\mathrm{Sym}^k(\mathbb{R}^d) = \left\{\xi : \mathbb{R}^d\times\cdots\times\mathbb{R}^d \mid \xi \text{ is } k\text{-linear and symmetric}\right\}.$$

Here, $C_c^k(\Omega,\mathrm{Sym}^k(\mathbb{R}^d))$ is the space of compactly supported symmetric tensor fields.

According to the [29], the TGV_α^k can be interpreted as a k-fold infimal convolution by employing the Fenchel–Rockafellar duality formula, which is formulated as follows:

$$\mathrm{TGV}_\alpha^k(\boldsymbol{U}) = \inf_{\boldsymbol{U}_j}\sum_{j=1}^{n}\alpha_{k-j}\int_\Omega |\mathcal{E}(\boldsymbol{U}_{j-1}) - \boldsymbol{U}_j|\,\mathrm{d}x, \tag{1}$$

where $\boldsymbol{U}_0 = \boldsymbol{U}, \boldsymbol{U}_k = 0, \boldsymbol{U}_j \in C_c^{k-j}\left(\Omega,\mathrm{Sym}^j(\mathbb{R}^d)\right)$ for $j = 1,\dots,k$, and $\mathcal{E}$ represents the distributional symmetrized derivative, i.e., $\mathcal{E}(\boldsymbol{U}_{j-1}) = \frac{\nabla\boldsymbol{U}_{j-1}+(\nabla\boldsymbol{U}_{j-1})^T}{2}$. Therefore, TGV_α^k automatically balances the first to the kth order derivatives of $\boldsymbol{U}$, effectively detecting edge features and reducing the staircase effect introduced by TV regularization.

Since HSI is a 3D data cube, the traditional TGV regularization cannot be directly applied. To address this, we extend the TGV regularization to 3D data, as detailed in Section 4.1.

4. HSI denoising using TGV regularized low-rank tensor decomposition

4.1. Proposed model

For a single-channel three-dimensional image $\mathcal{X} \in \mathbb{R}^{M\times N\times P}$, the discrete form of the TGV regularization expression, derived from (1), is given as follows:

$$\mathrm{TGV}(\mathcal{X}) = \min_{\mathcal{P}} \alpha_1\|\nabla\mathcal{X} - \mathcal{P}\|_1 + \alpha_0\|\mathcal{E}(\mathcal{P})\|_1, \tag{2}$$

Here, $\mathcal{P} = (\mathcal{P}_1,\mathcal{P}_2,\mathcal{P}_3)$ is the deformation field in the image space, and the operators for the image gradient $\nabla\mathcal{X}$ and the symmetric derivative $\mathcal{E}(\mathcal{P})$ are defined as follows:

$$\nabla\mathcal{X} = \left(\partial_x^+\mathcal{X}, \partial_y^+\mathcal{X}, \partial_z^+\mathcal{X}\right), \tag{3}$$

$$\mathcal{E}(\mathcal{P}) = \begin{pmatrix} \partial_x^-\mathcal{P}_1 & \frac{\partial_y^-\mathcal{P}_1+\partial_x^-\mathcal{P}_2}{2} & \frac{\partial_z^-\mathcal{P}_1+\partial_x^-\mathcal{P}_3}{2} \\ \frac{\partial_y^-\mathcal{P}_1+\partial_x^-\mathcal{P}_2}{2} & \partial_y^-\mathcal{P}_2 & \frac{\partial_z^-\mathcal{P}_2+\partial_y^-\mathcal{P}_3}{2} \\ \frac{\partial_z^-\mathcal{P}_1+\partial_x^-\mathcal{P}_3}{2} & \frac{\partial_z^-\mathcal{P}_2+\partial_y^-\mathcal{P}_3}{2} & \partial_z^-\mathcal{P}_3 \end{pmatrix}, \tag{4}$$

Here, ∂_x^+, ∂_y^+, and ∂_z^+ are the first-order forward difference, while ∂_x^-, ∂_y^-, and ∂_z^- are the first-order backward differences. Also, the divergence and Laplacian operators can be defined as follows:

$$\Delta\mathcal{X} = \mathrm{div}(\partial\mathcal{X}) = \partial_x^-\partial_x^+\mathcal{X} + \partial_y^-\partial_y^+\mathcal{X} + \partial_z^-\partial_z^+\mathcal{X}, \tag{5}$$

$$\mathrm{div}(\mathcal{P}) = \partial_x^-\mathcal{P}_1 + \partial_y^-\mathcal{P}_2 + \partial_z^-\mathcal{P}_3. \tag{6}$$

As previously mentioned, utilizing prior knowledge is a key consideration for removing mixed noise in HSIs. By combining the low-rankness and piecewise smoothness in the spatial–spectral dimension, we propose a 3D TGV regularized low-rank tensor decomposition model. Specifically, given a noise-corrupted observed HSI $\mathcal{Y} \in \mathbb{R}^{M\times N\times P}$, the proposed optimization model is formulated as follows:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{S},\mathcal{N}} \mathrm{TGV}(\mathcal{X}) + \lambda\|\mathcal{S}\|_1 + \beta\|\mathcal{N}\|_F^2, \\ &\text{s.t.}\quad \mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N}, \\ &\mathcal{X} = \mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3,\quad \mathbf{U}_i^T\mathbf{U}_i = \mathbf{I}\ (i=1,2,3). \end{aligned} \tag{7}$$

where $\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3$ represents the Tucker decomposition with core tensor $\mathcal{C}$ and factor matrices $\mathbf{U}_i$ of rank r_i. $\mathcal{X},\mathcal{S},\mathcal{N}$ have the same size of $\mathcal{Y}$, representing the clean data, sparse noise, and Gaussian noise.

Notably, the model can effectively remove mixed noise in the HSIs while well preserving detailed features. Specifically, Tucker decomposition captures the low-rank structural characteristics of HSIs by exploiting spatial–spectral correlations among pixels. Once the sparse noise is detected and separated by the ℓ_1-norm term, the TGV regularization terms work synergistically. The first-order term constrains the HSI's gradient, enforcing piecewise smoothness while preserving features and suppressing noise. Then, the second-order term constrains the HSI's spatial/spectral curvature and cross-band curvature, further promoting smoothness across the HSIs. Additionally, the Frobenius norm term models Gaussian noise and enhances the model's performance in scenarios with strong Gaussian noise interference.

It is evident that due to the non-convex nature of Tucker decomposition, the above problem is a non-convex optimization function. Therefore, in the next section, we will employ the ADMM to potentially find a good local solution.

4.2. Optimization procedure

In this section, we employ the ADMM to solve the minimization problem containing a TGV term in (7). First, we introduce some auxiliary variables $\mathcal{X} = \mathcal{Z}$, $\mathcal{F} = \nabla\mathcal{Z} - \mathcal{P}$, and $\mathcal{M} = \mathcal{E}(\mathcal{P})$, and rewrite (7) as the equivalent minimization problem:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{Z},\mathcal{S},\mathcal{N},\mathcal{P},\mathcal{F},\mathcal{M}} \alpha_1\|\mathcal{F}\|_1 + \alpha_0\|\mathcal{M}\|_1 + \lambda\|\mathcal{S}\|_1 + \beta\|\mathcal{N}\|_F^2 \\ &\text{s.t. } \mathcal{Y} = \mathcal{X}+\mathcal{S}+\mathcal{N}, \mathcal{X} = \mathcal{Z}, \mathcal{F} = \nabla\mathcal{Z} - \mathcal{P}, \mathcal{M} = \mathcal{E}(\mathcal{P}) \\ &\mathcal{X} = \mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3,\quad \mathbf{U}_i^T\mathbf{U}_i = \mathbf{I}\ (i=1,2,3). \end{aligned} \tag{8}$$

Based on the ADMM methodology, the above problem (8) can be transformed into minimizing the following augmented Lagrangian function:

$$\begin{aligned} &L(\mathcal{X},\mathcal{Z},\mathcal{S},\mathcal{N},\mathcal{P},\mathcal{F},\mathcal{M},\Gamma_1,\Gamma_2,\Gamma_3,\Gamma_4) = \alpha_1\|\mathcal{F}\|_1 + \alpha_0\|\mathcal{M}\|_1 + \\ &\lambda\|\mathcal{S}\|_1 + \beta\|\mathcal{N}\|_F^2 + \langle\Gamma_1,\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{N}\rangle + \langle\Gamma_2,\mathcal{X}-\mathcal{Z}\rangle + \langle\Gamma_3, \\ &\mathcal{F}-\nabla\mathcal{Z}+\mathcal{P}\rangle + \langle\Gamma_4,\mathcal{M}-\mathcal{E}(\mathcal{P})\rangle + \frac{\mu_1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{N}\|_F^2 + \frac{\mu_2}{2} \\ &\|\mathcal{X}-\mathcal{Z}\|_F^2 + \frac{\mu_3}{2}\|\mathcal{F}-\nabla\mathcal{Z}+\mathcal{P}\|_F^2 + \frac{\mu_4}{2}\|\mathcal{M}-\mathcal{E}(\mathcal{P})\|_F^2, \end{aligned} \tag{9}$$

under the constraints $\mathcal{X} = \mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3$ and $\mathbf{U}_i^T\mathbf{U}_i = \mathbf{I}\ (i=1,2,3)$, where Γ_i $(i=1,2,3,4)$ are Lagrange multipliers and μ_i $(i=1,2,3,4)$ are penalty parameters. Accordingly, we adopt an alternating optimization strategy for the augmented Lagrangian function (9), updating one variable at a time while keeping the others fixed. Specifically, during the $(k+1)$th iteration, the variables in model (7) are updated as follows:

(1) Update $\mathcal{C}, \mathbf{U}_i, \mathcal{X}$. The $\mathcal{X}$ subproblem is as follows:

$$\min_{\substack{\mathbf{U}_i^T\mathbf{U}_i=\mathbf{I},\\ \mathcal{X}=\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3}} \left\{ \langle \Gamma_1^k, \mathcal{Y}-\mathcal{X}-\mathcal{S}^k-\mathcal{N}^k\rangle + \langle \Gamma_2^k, \mathcal{X}-\mathcal{Z}^k\rangle + \frac{\mu_1}{2}\left\|\mathcal{Y}-\mathcal{X}-\mathcal{S}^k-\mathcal{N}^k\right\|_F^2 + \frac{\mu_2}{2}\left\|\mathcal{X}-\mathcal{Z}^k\right\|_F^2 \right\},$$

which can be easily transformed into the following equivalent problem:

$$\min_{\mathbf{U}_i^T\mathbf{U}_i=\mathbf{I}} \left\|\mathcal{C}\times_1\mathbf{U}_1\times_2\mathbf{U}_2\times_3\mathbf{U}_3 - \frac{1}{\mu_1+\mu_2}\left(\mu_1(\mathcal{Y}-\mathcal{S}^k-\mathcal{N}^k) + \mu_2\mathcal{Z}^k + (\Gamma_1^k-\Gamma_2^k)\right)\right\|_F^2.$$

By using the Higher-Order Orthogonal Iteration (HOOI) algorithm [58], we can easily get the $\mathcal{C}^{k+1}$, and $\boldsymbol{U}_i^{k+1}(i=1,2,3)$. Then $\mathcal{X}$ can be updated as follows:

$$\mathcal{X}^{k+1} = \mathcal{C}^{k+1}\times_1\mathbf{U}_1^{k+1}\times_2\mathbf{U}_2^{k+1}\times_3\mathbf{U}_3^{k+1}. \tag{10}$$

(2) Update $\mathcal{Z}$. The $\mathcal{Z}$ subproblem is as follows:

$$\mathcal{Z}^{k+1} = \underset{\mathcal{Z}}{\operatorname{argmin}}\left\{\langle \Gamma_2^k, \mathcal{X}^{k+1}-\mathcal{Z}\rangle + \langle \Gamma_3^k, \mathcal{F}^k-\nabla\mathcal{Z}+\mathcal{P}^k\rangle + \frac{\mu_2}{2}\|\mathcal{X}^{k+1}-\mathcal{Z}\|_F^2 + \frac{\mu_3}{2}\|\mathcal{F}^k-\nabla\mathcal{Z}+\mathcal{P}^k\|_F^2\right\}.$$

Thus, optimizing this subproblem can be treated as solving the following linear system:

$$(\mu_2\mathbf{I}+\mu_3\nabla^*\nabla)\mathcal{Z} = \nabla^*(\mu_3(\mathcal{F}^k+\mathcal{P}^k)+\Gamma_3^k)+\mu_2\mathcal{X}^{k+1}+\Gamma_2^k, \tag{11}$$

Here ∇^* indicates the adjoint operator of ∇. Due to the block-circulant structure of the matrix corresponding to the operator $\nabla^*\nabla$ [59], it can be diagonalized by the three-dimensional fast Fourier transform matrix. Therefore, (11) can be solved by

$$\mathcal{Z}^{k+1} = \operatorname{ifftn}\left(\frac{\operatorname{fftn}(\mu_2\mathcal{X}^{k+1}+\nabla^*(\mu_3(\mathcal{F}^k+\mathcal{P}^k)+\Gamma_3^k)+\Gamma_2^k)}{\mu_2\mathbf{I}+\mu_3(|\operatorname{fftn}(\nabla_x)|^2+|\operatorname{fftn}(\nabla_y)|^2+|\operatorname{fftn}(\nabla_z)|^2)}\right), \tag{12}$$

where fftn and ifftn indicate fast 3D Fourier transform and its inverse transform respectively. $\mathbf{I}$ denotes a tensor with all entries being 1. $|\cdot|^2$ is the elements-wise square, and the division is also performed element-wise.

(3) Update $\mathcal{P}$. The $\mathcal{P}$ subproblem is as follows:

$$\mathcal{P}^{k+1} = \underset{\mathcal{P}}{\operatorname{argmin}}\left\{\langle \Gamma_3, \mathcal{F}^k-\nabla\mathcal{Z}^{k+1}+\mathcal{P}\rangle + \langle \Gamma_4, \mathcal{M}^k-\mathcal{E}(\mathcal{P})\rangle + \frac{\mu_3}{2}\|\mathcal{F}^k-\nabla\mathcal{Z}^{k+1}+\mathcal{P}\|_F^2 + \frac{\mu_4}{2}\|\mathcal{M}^k-\mathcal{E}(\mathcal{P})\|_F^2\right\}. \tag{13}$$

For each $\mathcal{P}_1$, $\mathcal{P}_2$, and $\mathcal{P}_3$, there are three discrete formulations as follows:

$$\begin{aligned}
&(\mu_3-\mu_4\partial_x^+\partial_x^- - \frac{\mu_4}{2}\partial_y^+\partial_y^- - \frac{\mu_4}{2}\partial_z^+\partial_z^-)\mathcal{P}_1 - \frac{\mu_4}{2}\partial_y^+\partial_x^-\mathcal{P}_2 - \frac{\mu_4}{2}\partial_z^+\partial_x^-\mathcal{P}_3 = h_1,\\
&-\frac{\mu_4}{2}\partial_x^+\partial_y^-\mathcal{P}_1 + (\mu_3 - \frac{\mu_4}{2}\partial_x^+\partial_x^- - \mu_4\partial_y^+\partial_y^- - \frac{\mu_4}{2}\partial_z^+\partial_z^-)\mathcal{P}_2 - \frac{\mu_4}{2}\partial_z^+\partial_y^-\mathcal{P}_3 = h_2,\\
&-\frac{\mu_4}{2}\partial_x^+\partial_z^-\mathcal{P}_1 - \frac{\mu_4}{2}\partial_y^+\partial_z^-\mathcal{P}_2 + (\mu_3 - \frac{\mu_4}{2}\partial_x^+\partial_x^- - \frac{\mu_4}{2}\partial_y^+\partial_y^- - \mu_4\partial_z^+\partial_z^-)\mathcal{P}_3 = h_3.
\end{aligned} \tag{14}$$

where h_1, h_2, h_3 are expressed as follows:

$$\begin{aligned}
h_1 &= \mu_3(\nabla_x\mathcal{Z}^{k+1} - \mathcal{F}_1^k - \Gamma_{31}^k - \partial_x^+(\mu_4\mathcal{M}_1^k+\Gamma_{41}^k) - \partial_y^+(\mu_4\mathcal{M}_4^k + \Gamma_{44}^k) - \partial_z^+(\mu_4\mathcal{M}_5^k+\Gamma_{45}^k)),\\
h_2 &= \mu_3(\nabla_y\mathcal{Z}^{k+1} - \mathcal{F}_2^k - \Gamma_{32}^k - \partial_x^+(\mu_4\mathcal{M}_4^k+\Gamma_{44}^k) - \partial_y^+(\mu_4\mathcal{M}_2^k + \Gamma_{42}^k) - \partial_z^+(\mu_4\mathcal{M}_6^k+\Gamma_{46}^k)),\\
h_3 &= \mu_3(\nabla_z\mathcal{Z}^{k+1} - \mathcal{F}_3^k - \Gamma_{33}^k - \partial_x^+(\mu_4\mathcal{M}_5^k+\Gamma_{45}^k) - \partial_y^+(\mu_4\mathcal{M}_6^k + \Gamma_{46}^k) - \partial_z^+(\mu_4\mathcal{M}_3^k+\Gamma_{43}^k)).
\end{aligned} \tag{15}$$

By applying the discrete Fourier transform $\mathcal{F}$ to all sides of Eq. (14), we obtain the following linear system:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{pmatrix}\begin{pmatrix}\mathcal{F}(\mathcal{P}_1^{k+1})\\ \mathcal{F}(\mathcal{P}_2^{k+1})\\ \mathcal{F}(\mathcal{P}_3^{k+1})\end{pmatrix} = \begin{pmatrix}\mathcal{F}(h_1)\\ \mathcal{F}(h_2)\\ \mathcal{F}(h_3)\end{pmatrix}, \tag{16}$$

and easily obtain the analytical solution of $\mathcal{P}_1^{k+1}$, $\mathcal{P}_2^{k+1}$, and $\mathcal{P}_3^{k+1}$ with discrete inverse Fourier transform $\mathcal{F}^{-1}$.

(4) Update $\mathcal{F}$. The $\mathcal{F}$ subproblem is as follows:

$$\begin{aligned}
\mathcal{F}^{k+1} &= \underset{\mathcal{F}}{\operatorname{argmin}}\left\{\alpha_1\|\mathcal{F}\|_1 + \langle \Gamma_3^k, \mathcal{F}-\nabla\mathcal{Z}^{k+1}+\mathcal{P}^{k+1}\rangle + \frac{\mu_3}{2}\left\|\mathcal{F}-(\nabla\mathcal{Z}^{k+1}-\mathcal{P}^{k+1}-\Gamma_3^k)\right\|_F^2\right\},\\
&= \underset{\mathcal{F}}{\operatorname{argmin}}\left\{\alpha_1\|\mathcal{F}\|_1 + \frac{\mu_3}{2}\left\|\mathcal{F}-(\nabla\mathcal{Z}^{k+1}-\mathcal{P}^{k+1}-\frac{\Gamma_3^k}{\mu_3})\right\|_F^2\right\}.
\end{aligned}$$

By introducing the soft-thresholding operator, then we can update $\mathcal{F}^{k+1}$ as:

$$\mathcal{F}^{k+1} = \operatorname{Shrink}\left(\nabla\mathcal{Z}^{k+1}-\mathcal{P}^{k+1}-\frac{\Gamma_3^k}{\mu_3}, \frac{\alpha_1}{\mu_3}\right), \tag{17}$$

where $\operatorname{Shrink}(x,\tau) = \operatorname{sign}(x)\odot\max(|x|-\tau, 0)$ is the tensor soft shrinkage operator, $\odot$ denotes point-wise product.

(5) Update $\mathcal{M}$. The $\mathcal{M}$ subproblem is as follows:

$$\begin{aligned}
\mathcal{M}^{k+1} &= \underset{\mathcal{M}}{\operatorname{argmin}}\left\{\alpha_0\|\mathcal{M}\|_1 + \langle \Gamma_4^k, \mathcal{M}-\mathcal{E}(\mathcal{P}^{k+1})\rangle + \frac{\mu_4}{2}\|\mathcal{M}-\mathcal{E}(\mathcal{P}^{k+1})\|_F^2\right\},\\
&= \underset{\mathcal{M}}{\operatorname{argmin}}\left\{\alpha_0\|\mathcal{M}\|_1 + \frac{\mu_4}{2}\left\|\mathcal{M}-\left(\mathcal{E}(\mathcal{P}^{k+1})-\frac{\Gamma_4^k}{\mu_4}\right)\right\|_F^2\right\}.
\end{aligned}$$

Similar to the $\mathcal{F}$ problem, then the solution of above problem can be expressed as:

$$\mathcal{M}^{k+1} = \operatorname{Shrink}\left(\mathcal{E}(\mathcal{P}^{k+1})-\frac{\Gamma_4^k}{\mu_4}, \frac{\alpha_0}{\mu_4}\right). \tag{18}$$

(6) Update $\mathcal{S}$. The $\mathcal{S}$ subproblem is as follows:

$$\begin{aligned}
\mathcal{S}^{k+1} &= \underset{\mathcal{S}}{\operatorname{argmin}}\left\{\lambda\|\mathcal{S}\|_1 + \langle \Gamma_1^k, \mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}-\mathcal{N}^k\rangle + \frac{\mu_1}{2}\left\|\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}-\mathcal{N}^k\right\|_F^2\right\},\\
&= \underset{\mathcal{S}}{\operatorname{argmin}}\left\{\lambda\|\mathcal{S}\|_1 + \frac{\mu_1}{2}\left\|\mathcal{S}-\left(\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{N}^k+\frac{\Gamma_1^k}{\mu_1}\right)\right\|_F^2\right\}.
\end{aligned}$$

Similar to the $\mathcal{F}$ subproblem, then the solution of above problem can be expressed as:

$$\mathcal{S}^{k+1} = \operatorname{Shrink}\left(\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{N}^k+\frac{\Gamma_1^k}{\mu_1}, \frac{\lambda}{\mu_1}\right). \tag{19}$$

(7) Update $\mathcal{N}$. The $\mathcal{N}$ subproblem is as follows:

$$\begin{aligned}
\mathcal{N}^{k+1} &= \underset{\mathcal{N}}{\operatorname{argmin}}\left\{\beta\|\mathcal{N}\|_F^2 + \langle \Gamma_1^k, \mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}^{k+1}-\mathcal{N}\rangle + \frac{\mu_1}{2}\left\|\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}^{k+1}-\mathcal{N}\right\|_F^2\right\},\\
&= \underset{\mathcal{N}}{\operatorname{argmin}}\left\{(\beta+\frac{\mu_1}{2})\|\mathcal{N}\|_F^2 - \frac{\mu_1}{\mu_1+2\beta}\|\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}^{k+1}\|_F^2\right\}.
\end{aligned}$$

Thus through a simple calculation, one can get its solution as follows:

$$\mathcal{N}^{k+1} = \frac{\mu_1(\mathcal{Y}-\mathcal{X}^{k+1}-\mathcal{S}^{k+1})+\Gamma_1^k}{\mu_1+2\beta}. \tag{20}$$

(8) Update multipliers. Based on the ADMM algorithm, the multipliers are updated by the following equations:

$$\begin{cases} \Gamma_1^{k+1} = \Gamma_1^k + \mu_1(\mathcal{Y} - \mathcal{X}^{k+1} - \mathcal{S}^{k+1} - \mathcal{N}^{k+1}), \\ \Gamma_2^{k+1} = \Gamma_2^k + \mu_2(\mathcal{X}^{k+1} - \mathcal{Z}^{k+1}), \\ \Gamma_3^{k+1} = \Gamma_3^k + \mu_3(\mathcal{F}^{k+1} - \nabla \mathcal{Z}^{k+1} + \mathcal{P}^{k+1}), \\ \Gamma_4^{k+1} = \Gamma_4^k + \mu_4(\mathcal{M}^{k+1} - \mathcal{E}(\mathcal{P}^{k+1})). \end{cases} \tag{21}$$

Now we are ready to state the ADMM algorithm for solving problem (7) in Algorithm 1.

Algorithm 1 ADMM for solving model (7)

Input: The noisy HSI $\mathcal{Y}$, desired rank $[r_1, r_2, r_3]$, parameters $\alpha_1, \alpha_0, \lambda, \beta, \tau \in \left(0, \frac{(1+\sqrt{5})}{2}\right)$, and $\mu = [\mu_1, \mu_2, \mu_3, \mu_4]$.
Initialization: $\mathcal{X}^0 = 0, \mathcal{S}^0 = 0, \mathcal{N}^0 = 0, \mathcal{P}^0 = 0, \mathcal{M}^0 = 0, \mathcal{Z}^0 = 0, \Gamma_1 = \Gamma_2 = \Gamma_3 = \Gamma_4 = 0$ and $k_{\max}$.
while not converged and $k \le k_{\max}$ **do**
 Update $\mathcal{X}^{k+1}$ via (10);
 Update $\mathcal{Z}^{k+1}$ via (12);
 Update $\mathcal{P}^{k+1}$ via (16);
 Update $\mathcal{F}^{k+1}$ via (17);
 Update $\mathcal{M}^{k+1}$ via (18);
 Update $\mathcal{S}^{k+1}$ via (19);
 Update $\mathcal{N}^{k+1}$ via (20);
 Update $\Gamma_i^{k+1} (i = 1, ..., 4)$ via (21);
 Let $\mu_i = \tau\mu_i, (i = 1, 2, 3, 4)$;
 Check the stopping condition:

$$\frac{\|\mathcal{X}^{k+1} - \mathcal{X}^k\|_F}{\|\mathcal{X}^k\|_F} < 10^{-4} \text{ or } k > k_{\max};$$

end while
Output: The restored HSI $\mathcal{X}$.

5. Numerical experiments

In this section, we validate the performance of the proposed method using both simulated and real-world data. For the simulated experiments, we employ two datasets: a subimage of the Scene 5 dataset from the Hyperspectral Images of Natural Scenes 2002,[1] with a size of 300 × 300 × 31, and the Simu Indian dataset,[2] with a size of 145 × 145 × 224. The Scene 5 dataset contains complex spatial structures, making it suitable for evaluating spatial detail preservation, while the Simu Indian dataset is widely used in HSI denoising tasks due to its rich spectral content. These datasets jointly provide a balanced evaluation environment in terms of both spectral fidelity and spatial structure preservation. To further evaluate the effectiveness of our method on real-world noise, we conduct experiments on the WHU-Hi-HongHu dataset,[3] using a subimage with a size of 145 × 145 × 270, and the Indian Pines[2] dataset, with a size of 145 × 145 × 220. These real-world datasets contain naturally occurring noise and rich scene content, providing a more practical assessment of the algorithm's performance. In addition, Toys dataset,[4] with a size of 512 × 512 × 31, is used for multispectral image denoising, enabling a broader evaluation of the robustness and generalizability of our method across different spectral modalities. The pixel values of HSIs are normalized into [0, 1] band by-band. Eight related methods are employed for performance comparison with the proposed method: GoDec-based low-rank matrix recovery (LRMR) [47], spatial–spectral total variation regularization (SSTV) [20], low-rank tensor decomposition with total variation regularization (LRTDTV) [15], tensor nuclear norm-based methods, including 3DTNN and 3DlogTNN [60], robust tensor completion with TV regularization based on transformed tensor nuclear norm (TNTV) [61], supervised deep learning-based method (SDeCNN) [36], and 3 D spatial–spectral attention Transformer(TDSAT) [41]. All parameters are manually optimized according to the authors' given instructions. All experiments are implemented on MATLAB (R2020b) using Windows 11 with an AMD Ryzen 7 3845H 3.80 GHz and 32 GB RAM.

5.1. Simulated experiments

To assess the quality of the denoising results, we employ the mean peak signal-to-noise ratio (MPSNR) over all bands, the mean structural similarity index (MSSIM) over all bands, and the error relative global accuracy by synthesis (ERGAS). Generally, higher MPSNR, MSSIM, and lower ERGAS values are corresponding to better denoising.

The settings of the simulated noise applied to the Scene 5 and Simu Indian datasets are as follows:

Case 1: Gaussian Noise + Impulse Noise. Zero-mean Gaussian noise with a variance of 0.01 and impulse noise with a percentage of 0.2 were added to each band.

Case 2: Gaussian Noise + Impulse Noise + Deadlines. Zero-mean Gaussian noise with variance randomly selected in [0.01, 0.1], and impulse noise with percentage randomly selected in [0.1, 0.3], were added to each band. In addition, deadlines were added from band 201 to band 220 in the Simu Indian dataset and from band 21 to band 25 in the Scene 5 dataset. The number of stripes was randomly selected from [3, 10], and the stripe width was randomly generated within [1, 3].

Case 3: Gaussian Noise + Impulse Noise + Stripes. The Gaussian and impulse noise were added just as case 2. Then, stripes were added from band 71 to band 90 in Simu Indian and from band 13 to band 17 in Scene 5. The number of stripes was randomly selected between 20 and 40.

Case 4: Gaussian Noise + Impulse Noise + Deadlines + Stripes. Gaussian noise, impulse noise, and deadlines were added just as case 2. Then, stripes were added just as case 3, respectively.

Numerical comparison: Table 1 shows the MPSNR, MSSIM, and ERGAS results of nine competing methods on all datasets. The best and second-best results in each case are highlighted in bold and underline, respectively. As observed, the proposed method achieves superior results compared to the other methods in most cases. Particularly, for the Scene 5 dataset in case 2, we can see that the proposed method achieves around 2 dB gain in MPSNR beyond the second-best method because 3D TGV better identifies and preserves details. Furthermore, as shown in the spectral curves in Figs. 5 and 6, the proposed method significantly outperforms competing methods in terms of spectral fidelity. This is mainly attributed to the effective spatial–spectral continuity constraints imposed by the 3D TGV regularization, which enables the recovered spectral curves to more closely resemble those not polluted by noise.

Visual comparison: In Figs. 1 and 2, we present visual denoising results using different methods on the Scene 5 and Simu Indian datasets. Our results indicate that our method produces superior visual quality compared to competing methods. Specifically, our proposed method more effectively removes noise while preserving details. LRMR often retains residual Gaussian or impulse noise in various scenarios mainly because it models the local low-rank property of HSIs using patch-wise, while ignoring the commonly existing non local similarity structures in HSIs. SDeCNN tends to retain a significant amount of impulse noise as well. TDSAT produces over-smoothing restoration results. These two data-driven methods rely heavily on the diversity and size of the training dataset, and often exhibit poor generalization performance when applied to denoising tasks involving different HSI data. Although SSTV, LRTDTV, TNTV, 3DTNN, and 3DlogTNN can remove most of the

1 https://personalpages.manchester.ac.uk/staff/d.h.foster/default.html.
2 https://engineering.purdue.edu/∼biehl/MultiSpec/hyperspectral.html.
3 https://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm.
4 http://www1.cs.columbia.edu/CAVE/databases/multispectral.

Table 1
Numerical results of all compared methods on simulated experiments. Boldface and underline highlight the best and second-best values, respectively.

Data	Case	Metrics	Noisy	LRMR	SSTV	LRTDTV	TNTV	3DTNN	3DLogTNN	SDeCNN	TDSAT	Proposed
Scene 5	Case 1	MPSNR	11.315	33.656	36.958	39.467	36.656	39.588	40.101	21.215	32.164	**41.488**
		MSSIM	0.102	0.908	0.939	0.973	0.957	0.976	**0.985**	0.538	0.918	0.981
		ERGAS	1527.005	90.869	88.117	45.094	66.672	46.273	42.473	535.907	104.732	**41.057**
	Case 2	MPSNR	10.866	30.344	31.994	32.784	29.657	30.547	31.642	21.807	32.265	**34.746**
		MSSIM	0.089	0.830	0.834	0.875	0.792	0.891	0.894	0.557	0.916	**0.899**
		ERGAS	1521.225	128.298	143.783	106.634	161.951	109.609	123.858	471.679	103.598	**104.654**
	Case 3	MPSNR	11.262	31.255	33.480	33.069	31.491	32.390	31.374	21.121	31.897	**33.774**
		MSSIM	0.104	0.857	0.873	0.898	0.855	0.896	0.908	0.521	0.911	**0.909**
		ERGAS	1537.865	119.393	123.827	94.747	122.358	103.640	114.257	544.360	104.253	**90.560**
	Case 4	MPSNR	10.937	30.644	32.530	33.076	30.150	31.260	30.767	21.220	31.687	**33.091**
		MSSIM	0.096	0.840	0.849	0.879	0.813	0.881	0.882	0.529	0.908	**0.889**
		ERGAS	1647.119	127.378	139.318	104.466	153.538	117.991	123.444	485.817	104.547	**100.297**
Simu Indian	Case 1	MPSNR	12.251	39.467	44.463	45.679	42.893	43.716	42.028	25.884	24.562	**46.346**
		MSSIM	0.154	0.968	0.986	0.996	0.995	0.993	0.986	0.831	0.771	**0.997**
		ERGAS	572.582	45.031	18.200	14.929	19.720	16.127	19.925	126.973	145.722	**14.853**
	Case 2	MPSNR	12.145	34.621	38.879	39.850	37.408	37.445	37.468	25.683	24.568	**40.777**
		MSSIM	0.153	0.928	0.970	0.974	0.961	0.976	0.978	0.807	0.766	**0.983**
		ERGAS	594.093	76.999	36.333	27.573	52.211	78.898	63.811	132.711	146.158	**23.863**
	Case 3	MPSNR	12.134	34.560	39.724	39.109	37.400	37.331	37.978	25.637	24.564	**40.485**
		MSSIM	0.153	0.927	0.976	0.981	0.959	0.975	0.980	0.805	0.766	**0.985**
		ERGAS	594.182	76.054	32.110	28.158	51.054	76.827	65.383	133.072	146.322	**24.580**
	Case 4	MPSNR	12.123	34.780	38.570	39.528	37.384	37.563	37.762	25.639	24.561	**40.650**
		MSSIM	0.153	0.927	0.973	0.980	0.961	0.980	**0.984**	0.801	0.766	0.982
		ERGAS	595.312	76.225	40.184	**23.348**	59.433	77.642	62.656	133.286	146.539	24.088

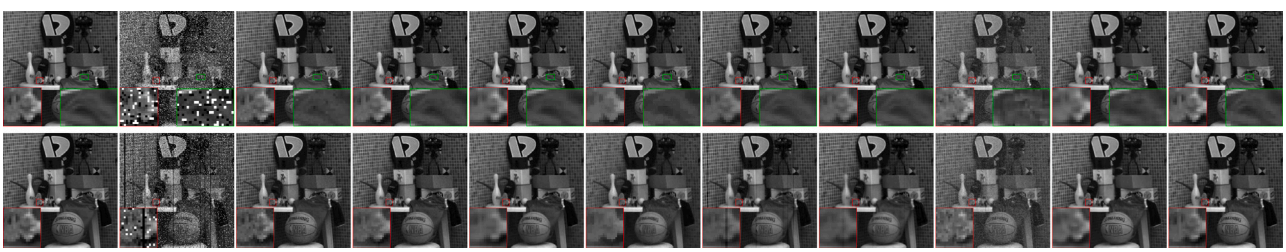


Fig. 1. Denoising results for Scene 5 in case 1 and case 4. The top row displays the visualizations of case 1 in the 24th band, while the bottom row presents case 4 in the 23td band.

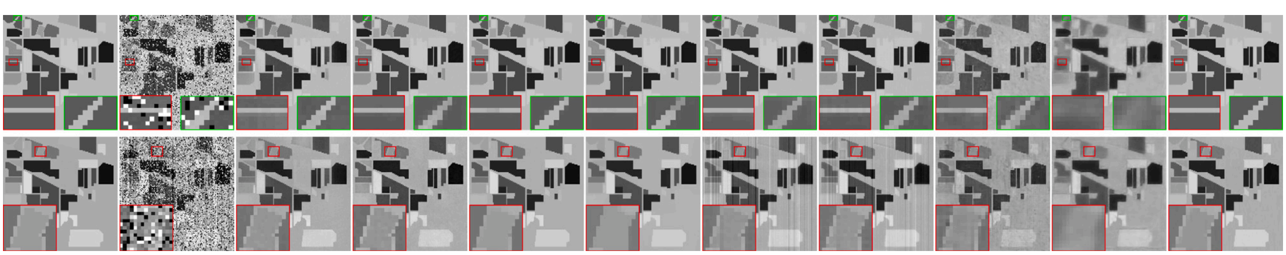


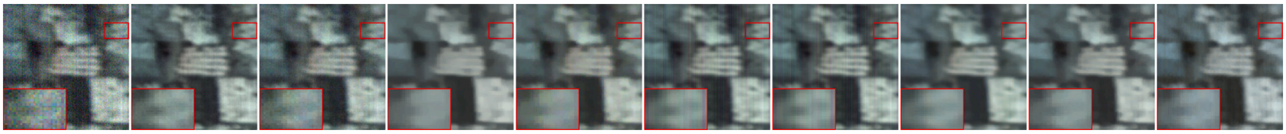
Fig. 2. Denoising results for Simu Indian in case 1 and case 3. The top row displays the visualizations of case 1 in the 70th band, while the bottom row presents case 3 in the 90td band.

noise, they exhibit noticeable over-smoothing in local detail regions, resulting in the loss of texture information and structural details. Notably, in Scene 5 case 1 and case 4, TNTV considers only first-order gradient information, which leads to noticeable staircase artifacts that are successfully avoided by our proposed method. Moreover, TNTV is more suitable for denoising HSIs with prominent edge structures. Additionally, when examining case 3 of Simu Indian and case 4 of Scene 5, the results from 3DlogTNN and 3DTNN exist noticeable residual stripes and deadlines, mainly due to the lack of structural modeling for such noise, which leads to their misidentification as valid structures.

5.2. Real experiments

In this subsection, since no ground-truth clean HSIs are available for quantitative comparison, the denoising performance was evaluated through visual results.

The WHU-Hi-HongHu dataset was polluted by severe Gaussian noise, stripes, and deadlines. The parameters of our proposed model were set as: $\alpha_1 = 0.2$, $\alpha_0 = 0.3$, $\lambda = 0.8$, $\beta = 30$, $\mu = [50, 50, 10, 1]$. Fig. 3 displays the visual results of WHU-Hi-HongHu, obtained using different methods. 3DTNN and 3DlogTNN results exhibit stripes,



Noisy LRMR SSTV LRTDTV TNTV 3DTNN 3DLogTNN SDeCNN TDSAT Proposed

Fig. 3. Pseudo-color recovered results (R: Band 246, G: Band 259, B: Band 270) of WHU-Hi-HongHu.

Noisy LRMR SSTV LRTDTV TNTV 3DTNN 3DLogTNN SDeCNN TDSAT Proposed

Fig. 4. Pseudo-color recovered results (R: Band 140, G: Band 110, B: Band 170) of AVIRIS.

in contrast to LRTDTV, SDeCNN, and TDSAT, which reduce noise but incur some detail loss. TNTV destroys texture details. LRMR and SSTV still retain some residual noise. In contrast, the proposed method effectively removes nearly all noise while better preserving details. The Indian Pines(AVIRIS) dataset was polluted by severe Gaussian noise and impulse noise. The parameters of our proposed model were set as: $\alpha_1 = 0.2$, $\alpha_0 = 0.3$, $\lambda = 1.3$, $\beta = 100$, $\mu = [50, 50, 10, 1]$. Fig. 4 shows visual results of AVIRIS by different methods. There exist noise in the results of LRMR, SSTV, 3DTNN, and 3DlogTNN, LRTDTV, TNTV, SDeCNN and TDSAT reduce noise but lose some details, while our method eliminates all of the noise while preserving details well.

5.3. Multispectral image experiments

In this subsection, we use Toys dataset to evaluate the performance of the proposed multispectral image denoising model. To validate the model's effectiveness, Gaussian noise, impulse noise, deadlines, and stripe noise are synthetically added. As shown in Fig. 7, LRMR, SSTV, and SDeCNN perform poorly on removing random noise, Although 3DTNN and 3DLogTNN are effective at suppressing random noise, they fail to eliminate coarse deadlines. LRTDTV, TNTV, and TDSAT perform well in removing mixed noise, while the restored results tend to be over-smoothing and cannot well preserve details. In comparison, our proposed method not only effectively suppresses various types of noise but also preserves details, demonstrating a favorable balance between noise removal and detail preservation. Moreover, it achieves significantly better performance metrics (see Table 2), with the MPSNR value exceeding the second-best result by 3.8 dB. These results demonstrate the robustness and generalizability of our method in handling complex noise conditions in multispectral images.

5.4. Discussions

(1) *Sensitivity analysis of parameters*. Here, we discuss the sensitivity of all parameters in the proposed method, including α_1, α_0, r, λ, and β. For parameters α_1 and α_0, we conduct a visual analysis to intuitively assess their impact on denoising, while a quantitative analysis of parameters λ, β, and rank r is performed in the case 4 of Scene 5 dataset. Parameter α_1 controls the impact of the first-order term in Eq. (8). For different noise scenarios, its reasonable range is [0.1, 1] (see Fig. 10(a)), leading to promising results (see Figs. 8(c) and 8(d)). If α_1 is too small, the first-order term is effectively ignored, resulting in $\mathcal{P} = 0$ in the HSI, which causes the second-order term to nearly vanish. Consequently, the TGV regularization fails, leaving residual noise in the results (see Fig. 8(b)). Conversely, if α_1 is too large, the first-order term tends to approach $\mathcal{P} = \nabla\mathcal{Z}$ near sharp features, and the minimization will be dominated by the second-order term in these areas, potentially smoothing out sharp features (see Fig. 8(e)).

Parameter α_0 affects the second-order term in Eq. (8). Similar to α_1, α_0 has a range of [0.1, 10] (see Fig. 10(b)) that can produce satisfactory results (see Figs. 9(c) and 9(d)). If the second-order term is insufficiently weighted, residual noise remains in the results (see Fig. 9(b)). In contrast, excessive weighting of the second-order term can damage smooth areas and fine details, resulting in over-smoothing of details (see Fig. 9(e)).

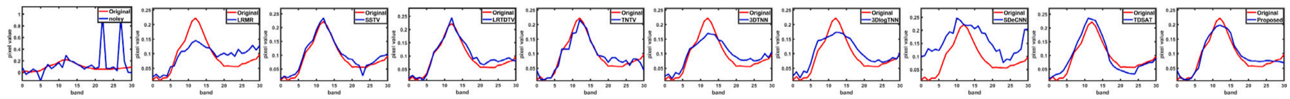
The rank r in Tucker decomposition represents the number of linearly independent components across the dimensions of the tensor, affecting the complexity of the decomposition and the accuracy of reconstruction. For different HSIs, we empirically adopt a rank size between 70% and 80% in the spatial dimension to leverage low-rank properties. Figs. 11(a) and 11(d) present the MPSNR and MSSIM values under different spectral mode rank constraints. It is evident that as the estimated rank increases, these values first rise and then quickly drop. The MPSNR and MSSIM values peaks at rank $r = 7$, indicating relative robustness within a certain range for rank r. Therefore, we select from the range [5, 10] to achieve the highest MPSNR and MSSIM values.

Parameter λ controls the sparse noise term. As shown in Figs. 11(b) and 11(e), with increasing values of λ, the MPSNR and MSSIM initially increase and then gradually decrease. This indicates that λ is relatively robust within a certain range. Thus, we select λ from the range [1, 3] to achieve the highest MPSNR and MSSIM values.

Parameter β controls the Gaussian noise term. Figs. 11(c) and 11(f) show the MPSNR and MSSIM values under different β values. It is observed that when β is slightly larger, the results remain robust. Therefore, in all experiments where the Gaussian noise level is not very high, we set β to 100. If the Gaussian noise level is high, the value of λ can be appropriately reduced.

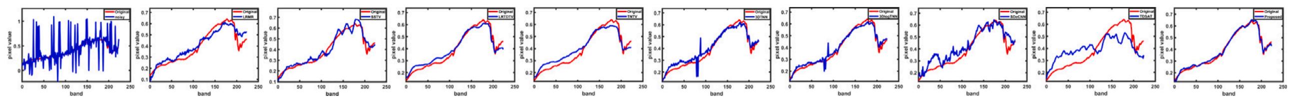
(2) *Convergence analysis*. In Figs. 12(a) and 12(b), we plot the curves of MPSNR and MSSIM gains with respect to iterations on Scene 5 in case 4. We can observe that, despite the non-convexity of proposed model, as the number of iterations increases to a relatively large value, the relative changes of MPSNR and MSSIM tend to zero. This clearly illustrates the strong convergence of the proposed method within the ADMM framework.

(3) *Model complexity analysis*. In Figs. 13(a) and 13(b) show the mean execution times of all competing models with varying noise levels for the Scene 5 and Simu Indian datasets,with specific runtime values provided. Notably, the two deep learning methods, SDeCNN and TDSAT, exhibited significantly better efficiencies than the seven traditional approaches. Owing to the increasing amount of computational data, LRMR, SSTV, LRTDTV, TNTV, 3DTNN, 3DlogTNN, and our method were relatively inefficient. This is reasonable because they adopt different dataexpansion schemes for data property enhancement. However, compared with TNTV, the execution efficiency of our method is acceptable. Furthermore, for compensation, our method reflects the low-rank property augmentation and the local spatial–spectral continuity.



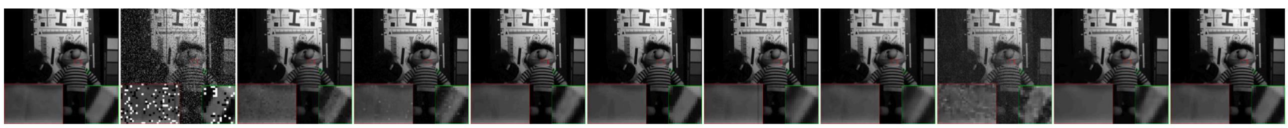
Noisy LRMR SSTV LRTDTV TNTV 3DTNN 3DLogTNN SDeCNN TDSAT Proposed

Fig. 5. Spectral curves for position (100, 180) of Scene 5 in case 4.



Noisy LRMR SSTV LRTDTV TNTV 3DTNN 3DLogTNN SDeCNN TDSAT Proposed

Fig. 6. Spectral curves for position (50, 40) of Simu Indian in case 3.



Clean Noisy LRMR SSTV LRTDTV TNTV 3DTNN 3DLogTNN SDeCNN TDSAT Proposed

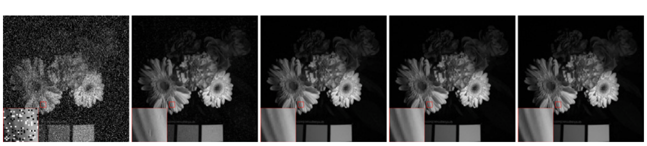
Fig. 7. Denoising results for the 27th Toys dataset with mixed noise (Gaussian noise, impulse noise, deadlines and stripes).

Table 2
Numerical results of all compared methods on Toys dataset. Boldface and underline highlight the best and second-best values, respectively.

Metrics	Noisy	LRMR	SSTV	LRTDTV	TNTV	3DTNN	3DLogTNN	SDeCNN	TDSAT	Proposed
MPSNR	10.676	34.939	34.681	<u>38.319</u>	36.007	30.057	32.469	19.547	32.618	**42.141**
MSSIM	0.165	0.918	0.936	0.973	<u>0.986</u>	0.980	0.982	0.430	0.906	**0.992**
ERGAS	1150.812	79.537	78.888	52.828	64.987	46.273	<u>44.382</u>	414.151	94.328	**32.008**

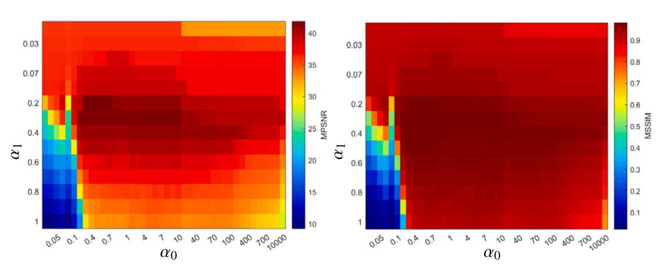
(a) Noisy (b) α_1=0.01 (c) α_1=0.3 (d) α_1=0.7 (e) α_1=1

Fig. 8. Denoising results for the 27th band of Flowers[4] dataset of size 512 × 512 × 31 by varying α_1 while keeping other parameters fixed.



(a) Noisy (b) α_0=0.05 (c) α_0=0.4 (d) α_0=5 (e) α_0=10

Fig. 9. Denoising results for the 27th band of Flowers dataset varying α_0 while keeping other parameters fixed.



(a) (b)

Fig. 10. Heatmap illustrating the effects of α_1 and α_0 on Flowers dataset recovery metrics: (a) MPSNR values, (b) MSSIM values.

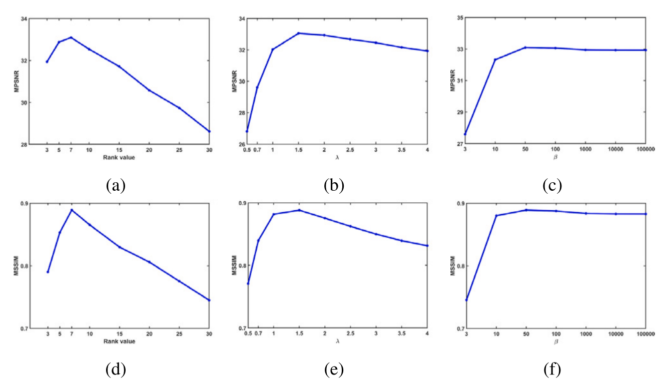
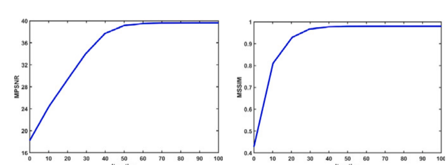


Fig. 11. MPSNR and MSSIM values with respect to different values of parameters r, λ, and β.

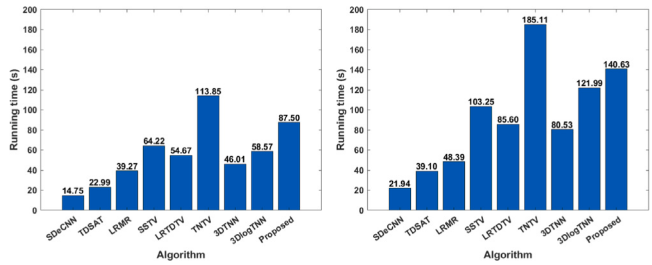


(a) (b)

Fig. 12. The changes in MPSNR and MSSIM values versus the iteration number of our proposed method.

6. Conclusion

For HSI mixed noise removal, this paper proposes a TGV regularized low-rank tensor decomposition model. The proposed method effectively removes noise while preserving fine details, reducing staircasing artifacts, and avoiding over-smoothing. To solve the model, we design



(a) (b)

Fig. 13. Execution time of all algorithms to recover HSIs: (a) Scene 5, (b) Simu Indian.

an efficient algorithm based on ADMM. A series of experiments with both simulated and real data demonstrate that our method outperforms other compared methods in both quantitative evaluation and visual comparison.

CRediT authorship contribution statement

Tingting Wei: Writing – original draft, Conceptualization. **Chunxue Wang:** Writing – review & editing. **Yajuan Li:** Supervision. **Chongyang Deng:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to thank Hongyan Zhang [47], Hemant Kumar Aggarwal [20], Yao Wang [15], Yubang Zheng [60], Duo Qiu [61], Alessandro Maffei [36], and Qiang Zhang [41] for sharing their code and the reviewers of this manuscript for their helpful comments and suggestions. This study was supported by the Hangzhou Dianzi University, China start-up fund (No. KYS075624218).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2025.104366.

Data availability

The data that has been used is confidential.

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